TELECOM CUSTOMER CHURN USING MACHINE LEARNING

Submitted in partial fulfilment of the requirements

for the award of

Bachelor of Engineering Degree in Computer Science and Engineering

by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF COMPUTING

SATHYABAMA

INSTITUTE OF SCIENCE AND TECHNOLOGY

(DEEMED TO BE UNIVERSITY)

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BONAFIDE CERTIFICATE

This is to certify that this Project Report is the bonafide work of **P. SAI PRAGNA (37110532) and B. ADWITHA (37110070)** who carried out the project entitled **"TELECOM CUSTOMER CHURN USING MACHINE LEARNING"** under our supervision from **November 2020** to **March 2021**.

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I **P.SAI PRAGNA** and **B.ADWITHA** hereby declare that the Project Report entitled "**TELECOM CUSTOMER CHURN USING MACHINE LEARNING**" done by us under the guidance of **Dr.L.Lakshmanan,M.E.,Ph.d**. is submitted in partial fulfillment of the requirements for the award of Bachelor ofComputer Science and Engineering.

DATE: 06-04-2021

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ABSTRACT

Service in the telecommunications industry is becoming one of the most difficult markets. The only communication with the industry has started to focus on technology in today's era, after the transition of cloud technology. If we have nothing, he adds, more pressure from the evils of things that exist in the industry, such as working in their housing business to date, to realign their crimpanies, as the examples will show. It takes in the plain of the lorem Intelligent machine servant of the increase of the thing as little. Survival to enumerate is not the key, as is the case, so that all Telcom companies better understand their customers and terminate them (if applicable) if they unsubscribe. With the right approach to employee attrition analysis, companies can identify and deliver their customer service, pricing, and marketing policies.

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CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

The telecommunications sector has become one of the main industries in developed countries. The technical progress and the increasing number of operators raised the level of competition. Companies are working hard to survive in this competitive market depending on multiple strategies. Three main strategies have been proposed to generate more revenues to acquire new customers, upsell the existing customers, and increase the retention period of customers. However, comparing these strategies taking the value of return on investment (ROI) of each into account has shown that the third strategy is the most profitable strategy, proves that retaining an existing customer costs much lower than acquiring a new one,in addition to being considered much easier than the upselling strategy. To apply the third strategy, companies have to decrease the potential of customer's churn, known as "the customer movement from one provider to another".

We focused on evaluating and analyzing the performance of a set of tree-based machine learning methods and algorithms for predicting churn in telecommunications companies. We have experimented a number of algorithms such as Decision Tree, Random Forest, Gradient Boost Machine Tree and to build the predictive model of customer Churn after developing our data preparation, feature engineering, and feature selection methods. Customers' churn is a considerable concern in service sectors with high competitive services. On the other hand, predicting the customers who are likely to leave the company will represent potentially large additional revenue source if it is done in the early phase.Many research confirmed that machine learning technology is highly efficient to predict this situation. This technique is applied through learning from previous data.

1.2 DOMAIN OVERVIEW

Machine learning is an application of Artificial intelligence (AI) that provides a systems the ability to automatically learn and improve from the experience without being explicitly programmed. Machine learning mainly focus on the development an computer logical programs that can access any data and use it learn for themselves.

The three major building blocks of a Machine Learning system are the model, the parameters, and the learner.

Model is the system which makes predictions.

The parameters are the factors which are considered by the model to make predictions.

The learner makes the adjustments in the parameters and the model to align the predictions with the actual results.

Fig 1.2 : Mechanism of machine

1.3 Block Diagram Of Machine

These are the seven steps that a machine learning project includes to give the best

Gathering Data

Preparing that data

Choosing a model

Training

Evaluation

Hyperparameter Tuning

Prediction

Formally, machine learning is the science of getting computers to realize a task without being explicitly programmed. In other words, the big difference between classical and machine learning algorithms lies in the way we define them.

Fig 1.3 : Simple Machine learning pipeline explanation

1.4 LITERATURE REVIEW

Churn : A churn is defined as customer attrition or loss in a telecom industry when customers terminate their contracts or their usage and switch to another service provider. There are type of customers that share important service features like higher bills, long distance transitions, etc. They spend more than the average rate and their expectations from the service providers is high, which make the retention of such customers more important than the rest as they are beneficial to the company. The company must allocate more resources to them to decrease the churn rate.

M.A.H. Farquad [4] proposed a hybrid approach to overcome the drawbacks of general SVM model which generates a black box model (i.e., it does not reveal the knowledge gained during training in human understandable form). The hybrid approach contains three phases: In the first phase, SVM-RFE (SVM-recursive feature elimination) is employed to reduce the feature set. In the second phase, dataset with reduced features is then used to obtain SVM model and support vectors are extracted. In the final phase, rules are then generated using Naive

Bayes Tree (NBTree which is combination of Decision tree with naive Bayesian Classifier). The dataset used here is bank credit card customer dataset (Business Intelligence Cup 2004) which is highly unbalanced with 93.24% loyal and 6.76% churned customers. The experimental showed that the model does not scalable to large datsets. datsets.

Ning Lu [7] proposed the use of boosting algorithms to enhance a customer churn prediction model in which customers are separated into two clusters based on the weight assigned by the boosting algorithm. As a result, a high risky customer cluster has been found. Logistic regression is used as a basis learner, and a churn prediction model is built on each cluster, respectively. The experimental results showed that boosting algorithm provides a good separation of churn data when compared with a single logistic regression model.

CHAPTER 2

AIM AND SCOPE OF PRESENT INVESTIGATION

2.1 AIM OF PROJECT

The aim of this post is to introduce a predictive model to Identify *the set of* customers who have a high probability of unsubscribing from the service now or in the near future using Personal Details, Demographic information, Pricing and the Plans they have subscribed to. A secondary objective is to identify the features of the Independent Variables (Predictors) which cause a great impact on the Dependent Variable (Y) that makes causes a customer to unsubscribe. So that the telecom companies can sustain their business by planning some strategies to maintain their customers without churning.

2.2 OBJECTIVES

- Exploratory analysis
- Data preparation
- Train, tune and evaluate machine learning models

2.3 SCOPE OF PROJECT

This document defines a unified architecture for Machine Learning in Fifth Generation and future networks. The unified logical architecture establishes a common vocabulary and nomenclature for ML functions and their interfaces. By applying (or superimposing) this logical architecture to a specific technology, like 3GPP or transport networks, the corresponding technology specific realization is derived. Such an overlay architecture for ML allows standardization and achieves interoperability for ML functions in Telecom Churn and future networks. In addition, the key issues for integration of such an overlay on Telecom Churn and future networks are studied and presented in this document.

What is a “Telecom customer churn”?

Customer churn, also known as customer attrition, occurs when customers stop doing business with a company or stop using a company's services. By being aware of and monitoring churn rate, companies are equipped to determine their customer retention success rates and identify strategies for improvement.Churn is a metric that shows customers who stop doing business with a company or a particular service, also known as customer attrition. By following this metric, what most businesses could do was try to understand the reason behind churn numbers and tackle those factors, with reactive action.But what if you could know in advance that a specific customer is likely to leave your business, and have a chance to take proper actions in time to prevent it from happening.The reasons that lead customers to the cancellation decision can be numerous, coming from poor service quality, delay on customer support, prices, new competitors entering the market, and so on. Usually, there is no single reason, but a combination of events that somehow culminated in customer displeasure.If your company were not capable to identify these signals and take actions prior to the cancel button click, there is no turning back, your customer is already gone. But you still have something valuable: the data. Your customer left very good clues about where you left to be desired. It can be a valuable source for meaningful insights and to train customer churn models. Learn from the past, and have strategic information at hand to improve future experiences, it’s all about machine learning.

When it comes to the telecommunications segment, there is great room for opportunities. The wealth and the amount of customer data that carriers collect can contribute a lot to shift from a reactive to a proactive position. The emergence of sophisticated artificial intelligence and data analytics techniques further help leverage this rich data to address churn in a much more effective manner.

As the purpose of this experiment is to identify patterns that can yield to customers churn, I will be focusing mainly on the churn portion of the dataset for the exploratory analysis.The customer lifespan (in months) is represented by the feature and customer churn by the churn feature , which is the target variable of this experiment. The bar chart below can provide right away a good insight on how churn is distributed across the customer lifespan.

Customer churn is the percentage of customers that stopped using your company's product or service during a certain time frame. You can calculate churn rate by dividing the number of customers you lost during that time period say a quarter by the number of customers you had at the beginning of that time period.

A churn model is a mathematical representation of how churn impacts your business. Churn calculations are built on existing data (the number of customers who left your service during a given time period). A predictive churn model extrapolates on this data to show future potential churn rates. Having the ability to accurate the model.

CHAPTER 3

SYSTEM DESIGN & METHODOLOGY

3.1 EXSISTING SYSTEM

In 4G, expanding interest for certain hardware or establishment doesn't need mandatory goal, the degree of information is improved, delays are decreased, and the nature of administration is higher. To address these issues, it is important to strengthen the advancement of rough versatile organizations. This record presents the aftereffects of broad examination into fifth era versatile organizations (SG) and explicit arising advancements that help improve the engineering and client needs. The collected data was full of columns, since there is a column for each service, product, and offer related to calls, SMS, MMS, and internet, in addition to columns related to personnel and demographic information. If we need to use all these data sources the number of columns for each customer before the data being processed will exceed ten thousand columns. Customer churn prediction has been performed using various techniques, including data mining, machine learning, and hybrid technologies. These techniques enable and support companies in identifying, predicting, and retaining churn customers. They also help industries in CRM and decision making. Most of them used decision trees in common as it is one of the recognized methods to find out the customer churn, but it is not appropriate for complex problems. But the study shows that reducing the data improves the accuracy of the decision tree.In some cases, data mining algorithms are used for customer prediction and historical analysis.

The techniques of regression trees were discussed with other commonly used data mining methods like decision trees, rule-based learning, and neural networks.Gavril et al presented an advanced methodology of data mining to predict churn for prepaid customers using dataset for call details of 3333 customers with 21 features, and a dependent churn parameter with two values: Yes/No. Some features include information about the number of incoming and outgoing messages and voicemail for each customer. The author applied principal component analysis algorithm “PCA” to reduce data dimensions. Three machine learning algorithms were used: Neural Networks, Support Vector Machine, and Bayes Networks to predict churn factor. The author used AUC to measure the performance of the algorithms. The AUC values were 99.10%, 99.55% and 99.70% for Bayes Networks, Neural networks and support vector machine, respectively. The dataset used in this study is small and no missing values existed.Various researches studied the problem of unbalanced data sets where the churned customer classes are smaller than the active customer classes, as it is a major issue in churn prediction problem. Amin et al compared six different sampling techniques for oversampling regarding telecom churn prediction problem. The results showed that the algorithms (MTDF and rules-generation based on genetic algorithms) outperformed the other compared oversampling algorithms.

DRAWBACKS

- Old devices can't support this- Hardware replacement will be required.
- At present, 5G technology is theoretical, it may take longer than expected to come into existence.
- Logistic regression is not able to handle a large number of categorical features/variables.
- It can't solve the non-linear problem with the logistic regression that is why it requires a transformation of non-linear features.
- It is vulnerable to overfitting.

3.2 PROPOSED SYSTEM

Customer in purchase price (CAC) constituting marketing, advertising, sales expenses, etc. And almost the same as 5 times the cost of customer loyalty value (C RC). In this way, the value of a price in the house of customer of clients is very important there. The model for prediction based on the Neural Network algorithm in order to solve the problem of customer churn in a large Chinese telecom company which contains about 5.23 million customers. The prediction accuracy standard was the overall accuracy rate, and reached 91.1%. In a business scenario predicting customer churn is where a firm is attempting a retain customer which is much probable to leave the services.For reducing the rate of churn this study classifies which customers are much going to churn probably and which will not churn probably. Since obtaining new customers is challenging it is essential to retain present customers. Churn can be decreased by examining the essential customers past history systematically. Huge amount of data is managed about the customers and on carrying out appropriate examination on the same it is feasible to find probable customers that might churn.

The data that is feasible can be examined in varied ways and thereby offers different ways for operators to imagine the churning of customers and avoid the same. The below figure shows the steps used for proposed system. In this system, we use various algorithms like Random Forest, XGBoost & Logistic Regression to find accurate values and which helps us to predict the churn of the customer. Here we implement the model by having a dataset that is trained and tested, which makes us have maximum correct values. Fig.1 shows the proposed model for churn prediction and describes its steps. In the Initial step, data preprocessing is performed in which we do filtering data and convert data into a similar form, and then we make feature selection.In the further step prediction and classification is done using the algorithms like Random Forest, XGBoost, Logistic Regression(LR). Training and testing the model with the data set, we observe the behaviour of the customer and analyze them. In the final step, we do analysis based on the results obtained and predict the customer churn.

ADVANTAGES OF PROPOSED SYSTEM

- It has the ability to accurately predict future churn.
- It helps your business gain a better understanding of future expected revenue.
- Predicting churn rates can also help your business identify and improve upon areas where customer service is lacking.
- Computationally efficient.
- These networks can learn from examples and apply them when a similar event arises,making them able to work through real-time events.

3.3 REQUIREMENT SPECIFICATION

Clearly defined requirements are essential signs on the road that leads to a successful project. They establish a formal agreement between a client and a provider that they are both working to reach the same goal. High-quality, detailed requirements also help mitigate financial risks and keep the project on a schedule.

According to the Business Analysis Body of Knowledge definition, requirements are a usable representation of a need. Creating requirements is a complex task as it includes a set of processes such as elicitation, analysis, specification, validation, and management. In this article, we'll discuss the main types of requirements for software products and provide a number of recommendations for their use.The requirements are split into two categories namely:

Software Requirements

The basic software requires to rum the program are :

Software	**Jupyter Notebook**
Language	**Python**
IDE	**Eclipse**
Front-end	**HTML , CSS**
Browser	**Chrome**

Hardware Requirements

The basic hardware required to run the program are :

System	**DELL CORE i3**
Hard Disk	**256 GB**
Monitor	**15" LED**
Input Devices	**Mouse , Keyboard**
RAM	**8 GB**

3.4 SOFTWARE DESCRIPTION

3.4.1 PYTHON

Python is a free, open-source programming language. Therefore, all you have to do is install Python once, and you can start working with it. Not to mention that you can contribute your own code to the community. Python is also a cross-platform compatible language. So, what does this mean? Well, you can install and run Python on several operating systems.

Python is also a great visualization tool. It provides libraries such as Matplotlib, seaborn and bokeh to create stunning visualizations.In addition, Python is the most popular language for machine learning and deep learning. As a matter of fact, today, all top organizations are investing in Python to implement machine learning in the back-end.

3.4.2 NumPy

NumPy, which stands for Numerical Python, is a library consisting of multidimensional array objects and a collection of routines for processing those arrays. Using NumPy, mathematical and logical operations on arrays can be performed.NumPy is a Python package. It stands for 'Numerical Python'. It is a library consisting of multidimensional array objects and a collection of routines for processing of array.Numeric, the ancestor of NumPy, was developed by Jim Hugunin. Another package Numarray was also developed, having some additional functionalities. In 2005, Travis Oliphant created NumPy package by incorporating the features of Numarray into Numeric package. There are many contributors to this open source project.

Operations using NumPy, a developer can perform the following operations –

1.Mathematical and logical operations on arrays.

2. Fourier transforms and routines for shape manipulation.

3. Operations related to linear algebra. NumPy has in-built functions for linear algebra and random number generation.

3.4.3 Pandas

Pandas is defined as an open-source library that provides high-performance data manipulation in Python. The name of Pandas is derived from the word Panel Data, which means an Econometrics from Multidimensional data. It is used for data analysis in Python and developed by Wes McKinney in 2008.Data analysis requires lots of processing, such as restructuring, cleaning or merging, etc. There are different tools are available for fast data processing, such as Numpy, Scipy, Cython, and Panda. But we prefer Pandas because working with Pandas is fast, simple and more expressive than other tools.Pandas is built on top of the Numpy package, means Numpy is required for operating the Pandas.Before Pandas, Python was capable for data preparation, but it only provided limited support for data analysis. So, Pandas came into the picture and enhanced the capabilities of data analysis. It can perform five significant steps required for processing and analysis of data irrespective of the origin of the data, i.e., load, manipulate, prepare, model, and analyze.

Key features of Pandas :

1. It has a fast and efficient DataFrame object with the default and customized

indexing.

2. Used for reshaping and pivoting of the data sets.

3. Group by data for aggregations and transformations.

4. It is used for data alignment and integration of the missing data. 5.Provide the functionality of Time Series.

6. Process a variety of data sets in different formats like matrix data, tabular heterogeneous, time series.

7. Handle multiple operations of the data sets such as subsetting, slicing, filtering, groupBy, re-ordering, and re-shaping.

8. It integrates with the other libraries such as SciPy, and scikit-learn.

9. Provides fast performance, and If you want to speed it, even more, you can use the Cython.

Benefits of Pandas :

The benefits of pandas over using other language are as follows:

1. Data Representation: It represents the data in a form that is suited for data analysis through its DataFrame and Series.

2. Clear code: The clear API of the Pandas allows you to focus on the core part of the code. So, it provides clear and concise code for the user.

3.4.4 Matplotlib

Matplotlib is one of the most popular Python packages used for data visualization. It is a cross-platform library for making 2D plots from data in arrays. Matplotlib is written in Python and makes use of NumPy, the numerical mathematics extension of Python. It provides an object-oriented API that helps in embedding plots in applications using Python GUI toolkits such as PyQt, WxPythonotTkinter. It can be used in Python and IPython shells, Jupyter notebook and web application servers also.

Matplotlib has a procedural interface named the Pylab, which is designed to resemble MATLAB, a proprietary programming language developed by MathWorks. Matplotlib along with NumPy can be considered as the open source equivalent of MATLAB.

Matplotlib is a Python Library used for plotting, this python library provides and objected-oriented APIs for integrating plots into applications. There are various plots which can be created using python matplotlib. Some of them are listed below:

3.4.5 SeaBorn

Seaborn is a library for making statistical graphics in Python. Seaborn helps you explore and understand your data. Its plotting functions operate on dataframes and arrays containing whole datasets and internally perform the necessary semantic mapping and statistical aggregation to produce informative plots. Seaborn is a library that uses Matplotlib underneath to plot graphs. It will be used to visualize random distributions.

Relationship to matplotlib:

Seaborn's integration with matplotlib allows you to use it across the many environments that matplotlib supports, inlcuding exploratory analysis in notebooks, real-time interaction in GUI applications, and archival output in a number of raster and vector formats. While you can be productive using only seaborn functions, full customization of your graphics will require some knowledge of matplotlib’s concepts and API. One aspect of the learning curve for new users of seaborn will be knowing when dropping down to the matplotlib layer is necessary to achieve a particular customization. On the other hand, users coming from matplotlib will find that much of their knowledge transfers.

Matplotlib has a comprehensive and powerful API; just about any attribute of the figure can be changed to your liking. A combination of seaborn's high-level interface and matplotlib's deep customizability will allow you both to quickly explore your data and to create graphics that can be tailored into a publication quality final product.

3.4.6 Flask

Flask is a micro web framework written in Python. It is classified as a microframework because it does not require particular tools or libraries. It has no database abstraction layer, form validation, or any other components where pre-existing third-party libraries provide common functions. However, Flask supports extensions that can add application features as if they were implemented in Flask itself. Extensions exist for object-relational mappers, form validation, upload handling, various open authentication technologies and several common framework related tools.

Flask is a popular Python web framework, meaning it is a third-party Python library used for developing web applications. Applications that use the Flask framework include Pinterest and LinkedIn.

3.4.7 Chrome

Google Chrome is a cross-platform web browser developed by Google. It was first released in 2008 for Microsoft Windows, and was later ported to Linux, macOS, iOS, and Android where it is the default browser built into the OS. The browser is also the main component of Chrome OS, where it serves as the platform for web applications.

Most of Chrome's source code comes from Google's free and open-source software project Chromium, but Chrome is licensed as proprietary freeware. WebKit was the original rendering engine, but Google eventually forked it to create the Blink engine; all Chrome variants except iOS now use Blink. As of November 2020, StatCounter estimates that Chrome has a 70% worldwide browser market share (after peaking at 72.38% in November 2018) on personal computers (PC), and 66.12% across all platforms. Because of this success, Google has expanded the "Chrome" brand name to other products: Chrome OS, Chromecast, Chromebook, Chromebit, Chromebox, and Chromebase.

3.4.8 Eclipse

The Eclipse IDE is famous for our Java Integrated Development Environment (IDE), but we have a number of pretty cool IDEs, including our C/C++ IDE, JavaScript/TypeScript IDE, PHP IDE, and more. Developed using Java, the Eclipse platform can be used to develop rich client applications, integrated development environments and other tools. Eclipse can be used as an IDE for any programming language for which a plug-in is available.

Eclipse is also good for a beginner. Its free, has a ton of features, and its straight-up Java features are great.

CHAPTER 4

SOFTWARE DEVELOPMENT METHODOLOGY

4.1 DESCRIPTION OF DIAGRAM

The machine learning architecture defines the various layers involved in the machine learning cycle and involves the major steps being carried out in the transformation of raw data into training data sets capable for enabling the decision making of a system. It is mainly used for optimizing the available resources and output based on the historical data available, additionally, machine learning involves major advantages about data forecasting and predictive analytics when coupled with data science technology.

Fig 4.1 : System Architecture

It involves various steps involved in the process of predicting whether a customer is getting churned or not.The process includes various modules like preparing the dataset for processing the algorithm on it and prepare a model to predict the class so that it becomes easy for humans like us to take the measures regarding the problem.

The architecture of this particular project consists of the following modules:

- Data Pre-processing
- Data Analysis
- Feature engineering
- Model selection
- Optimized algorithm

4.2 SYSTEM IMPLEMENTATION

The implementation starts with a dataset collection. The dataset thet we considered in our project is collected from online which consists of data from all the states of U.S.A .It consists all the data corresponding the telecom services like day_calls, eve_calls ,night calls, state name, plan name, the number of times the customer called the customer care and the number of times they contacted the customer care through messgaes ,etc., these are some of the features that are present in the dataset and there are total of 21 features like this and 5000 entries of customers.

4.2.1 Data Pre-processing:

Data preprocessing in Machine Learning is a crucial step that helps enhance the quality of data to promote the extraction of meaningful insights from the data. Data preprocessing in Machine Learning refers to the technique of preparing (cleaning and organizing) the raw data to make it suitable for a building and training Machine Learning models. In simple words, data preprocessing in Machine Learning is a data mining technique that transforms raw data into an understandable and readable format.

When it comes to creating a Machine Learning model, data preprocessing is the first step marking the initiation of the process. Typically, real-world data is incomplete, inconsistent, inaccurate (contains errors or outliers), and often lacks specific attribute values/trends. This is where data preprocessing enters the scenario – it helps to clean, format, and organize the raw data, thereby making it ready-to-go for Machine Learning models.

One Hot Encoding

One hot encoding is a process by which categorical variables are converted into a form that could be provided to ML algorithms to do a better job in prediction. One Hot Encoding is a common way of preprocessing categorical features for machine learning models. This type of encoding creates a new binary feature for each possible category and assigns a value of 1 to the feature of each sample that corresponds to its original category. It's easier to understand visually: in the example below, we One Hot Encode a color feature which consists of three categories (red, green, and blue).

4.2.2 Data Analysis

Before feeding your labeled data to an ML algorithm, it is a good practice to inspect your data to identify issues and gain insights about the data you are using. The predictive power of your model will only be as good as the data you feed it.

When analyzing your data, you should keep the following considerations in mind:

Variable and target data summaries – It is useful to understand the values that your variables take and which values are dominant in your data. You could run these summaries by a subject matter expert for the problem that you want to solve. Ask yourself or the subject matter expert: Does the data match your expectations? Does it look like you have a data collection problem? Is one class in your target more frequent than the other classes? Are there more missing values or invalid data than you expect?

Variable-target correlations – Knowing the correlation between each variable and the target class is helpful because a high correlation implies that there is a relationship between the variable and the target class. In general, you want to include variables with high correlation because they are the ones with higher predictive power (signal), and leave out variables with low correlation because they are likely irrelevant.

EDA with visualization

Exploratory Data Analysis (EDA) is an approach for data analysis that employs a variety of techniques to maximize insights into a dataset; uncover the underlying structure; extract important variables; detect outliers and anomalies; develop parsimonious models; and determine optimal factor settings.

The EDA types of techniques are either graphical or quantitative (non-graphical). While the graphical methods involve summarising the data in a diagrammatic or visual way, the quantitative method, on the other hand, involves the calculation of summary statistics. These two types of methods are further divided into univariate and multivariate methods. Univariate methods consider one variable (data column) at a time, while multivariate methods consider two or more variables at a time to explore relationships. Thus, there are four types of EDA in all — univariate graphical, multivariate graphical, univariate non-graphical, and multivariate non-graphical. The graphical methods provide more subjective analysis, and quantitative methods are more objective.

The objectives of EDA can be summarized as follows:

1. Maximize insight into the database/understand the database structure;
2. Visualize potential relationships (direction and magnitude) between exposure and outcome variables;
3. Detect outliers and anomalies (values that are significantly different from the other observations);
4. Develop parsimonious models (a predictive or explanatory model that performs with as few exposure variables as possible) or preliminary selection of appropriate models;
5. Extract and create clinically relevant variables.

Understanding Your Variables, You don't know what you don't know. Cleaning your dataset , Analyzing relationships between variables.

These are the main three objectives of EDA to make the dataset clear.

4.3 Feature Engineering

Each feature, or column, represents a measurable piece of data that can be used for analysis: Name, Age, Fare, and so on. Features are also sometimes referred to as "variables" or "attributes." Depending on what you're trying to analyze, the features you include in your dataset can vary widely.

Basically, all machine learning algorithms use some input data to create outputs. This input data comprise features, which are usually in the form of structured columns. Algorithms require features with some specific characteristic to work properly. Here, the need for feature engineering arises. I think feature engineering efforts mainly have two goals :

- Preparing the proper input dataset, compatible with the machine learning algorithm requirements.
- Improving the performance of machine learning models

The features you use influence more than everything else the result. No algorithm alone, to my knowledge, can supplement the information gain given by correct feature engineering.

Missing values are one of the most common problems you can encounter when you try to prepare your data for machine learning. The reason for the missing values might be human errors, interruptions in the data flow, privacy concerns, and so on. Whatever is the reason, missing values affect the performance of the machine learning models.Some machine learning platforms automatically drop the rows which include missing values in the model training phase and it decreases the model performance because of the reduced training size. On the other hand, most of the algorithms do not accept datasets with missing values and gives an error.

The most simple solution to the missing values is to drop the rows or the entire column. There is not an optimum threshold for dropping but you can use 70% as an example value and try to drop the rows and columns which have missing values with higher than this threshold.

SCALING: Feature Scaling is a technique to standardize the independent features present in the data in a fixed range. It is performed during the data pre-processing to handle highly varying magnitudes or values or units. If feature scaling is not done, then a machine learning algorithm tends to weigh greater values, higher and consider smaller values as the lower values, regardless of the unit of the values.

Example: If an algorithm is not using feature scaling method then it can consider the value 3000 meter to be greater than 5 km but that's actually not true and in this case, the algorithm will give wrong predictions. So, we use Feature Scaling to bring all values to same magnitudes and thus, tackle this issue.

4.4 MODEL SELECTION

Model selection is the process of selecting one final machine learning model from among a collection of candidate machine learning models for a training dataset.Model selection is a process that can be applied both across different types of models (e.g. logistic regression, SVM, KNN, etc.) and across models of the same type configured with different model hyperparameters (e.g. different kernels in an SVM). Model selection is the process of selecting one final machine learning model from among a collection of candidate machine learning models for a training dataset.

Model selection is the process of choosing one of the models as the final model that addresses the problem.Model selection is different from model assessment. All models have some predictive error, given the statistical noise in the data, the incompleteness of the data sample, and the limitations of each different model type. Therefore, the notion of a perfect or best model is not useful. Instead, we must seek a model that is "good enough."

SVM algorithm:

Support Vector Machine" (SVM) is a supervised machine learning algorithm which can be used for both classification or regression challenges. However, it is mostly used in classification problems. In the SVM algorithm, we plot each data item as a point in n-dimensional space (where n is number of features you have) with the value of each feature being the value of a particular coordinate. Then, we perform classification by finding the hyper-plane that differentiates the two classes very well.

The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future. This best decision boundary is called a hyperplane. SVM chooses the extreme points/vectors that help in creating the hyperplane. These extreme cases are called as support vectors, and hence algorithm is termed as Support Vector Machine. Consider the below diagram in which there are two different categories that are classified using a decision boundary or hyperplane.

Linear SVM: The working of the SVM algorithm can be understood by using an example. Suppose we have a dataset that has two tags (green and blue), and the dataset has two features x1 and x2. We want a classifier that can classify the pair(x1, x2) of coordinates in either green or blue. So as it is 2-d space so by just using a straight line, we can easily separate these two classes. But there can be multiple lines that can separate these classes. Hence, the SVM algorithm helps to find the best line or decision boundary; this best boundary or region is called as a hyperplane. SVM algorithm finds the closest point of the lines from both the classes. These points are called support vectors. The distance between the vectors and the hyperplane is called as margin. And the goal of SVM is to maximize this margin. The hyperplane with maximum margin is called the optimal hyperplane.

LOGISTIC REGRESSION:

Logistic regression is a classification algorithm used to assign observations to a discrete set of classes. Some of the examples of classification problems are Email spam or not spam, Online transactions Fraud or not Fraud, Tumor Malignant or Benign. Logistic regression transforms its output using the logistic sigmoid function to return a probability value.

There are two types:

Binary (eg. Tumor Malignant or Benign)

Multi-linear functions failsClass (eg. Cats, dogs or Sheep's)

We can call a Logistic Regression a Linear Regression model but the Logistic Regression uses a more complex cost function, this cost function can be defined as the 'Sigmoid function' or also known as the 'logistic function' instead of a linear function. In order to map predicted values to probabilities, we use the Sigmoid function. The function maps any real value into another value between 0 and 1. In machine learning, we use sigmoid to map predictions to probabilities.

Before diving into the implementation of logistic regression, we must be aware of the following assumptions about the same In case of binary logistic regression, the target variables must be binary always and the desired outcome is represented by the factor level 1.There should not be any multi-collinearity in the model, which means the independent variables must be independent of each other.We must include meaningful variables in our model.We should choose a large sample size for logistic regression.

RANDOM FOREST ALGORITHM:

Random forest is a supervised learning algorithm which is used for both classification as well as regression. But however, it is mainly used for classification problems. As we know that a forest is made up of trees and more trees means more robust forest. Similarly, random forest algorithm creates decision trees on data samples and then gets the prediction from each of them and finally selects the best solution by means of voting. It is an ensemble method which is better than a single decision tree because it reduces the over-fitting by averaging the result.

We can understand the working of Random Forest algorithm with the help of following steps −

Step 1 − First, start with the selection of random samples from a given dataset.

Step 2 − Next, this algorithm will construct a decision tree for every sample. Then it will get the prediction result from every decision tree.

Step 3 − In this step, voting will be performed for every predicted result.

Step 4 − At last, select the most voted prediction result as the final prediction result.

Fig 4.4 : Mechanism of Random Forest Algorithm

In this way random forest algorithm works and gives us the best result in classifying the class and predicting the accuracy percentage.

CHAPTER 5

RESULTS AND DISCUSSIONS

Fig 5.1 : Accuracy score of Logistic Regression

In the above picture it represents the analysis don using Logistic regression algorithm and we got an accuracy of 78% which is very less than the other two algorithms.

Fig 5.2 : Accuracy score of SVM

In the above picture it is representing the analysis of SVM algorithm where it gave an accuracy of 87%.

Fig 5.3 : Accuracy score of Random Forest algorithm

In the above picture it is representing the analysis of Random forest algorithm where it is giving the highest accuracy of 94% in predicting about the churn and non-churn.

Fig 5.4 : Displaying Risk analysis in front end

In the above picture it is representing the user end information. When we select a entry it is showing as "At risk" which means that particular customer is going to churn.

Fig 5.5 : Displaying Risk analysis in front end

In the above picture it is representing the user end information. When we select a entry it is showing as “Not at risk” which means that particular customer is not going to churn.

CHAPTER 6

SUMMARY AND CONCLUSION

6.1 CONCLUSION

The importance of this type of research in the telecom market is to help companies make more profit. It has become known that predicting churn is one of the most important sources of income to telecom companies. Hence, this research aimed to build a system that predicts the churn of customers in telecom company. These prediction models need to achieve high AUC values. To test and train the model, the sample data is divided into 70% for training and 30% for testing. We chose to perform cross-validation with 10-folds for validation and hyperparameter optimization. We have applied feature engineering, effective feature transformation and selection approach to make the features ready for machine learning algorithms. In addition, we encountered another problem: the data was not balanced. Only about 5% of the entries represent customers' churn. This problem was solved by undersampling or using trees algorithms not affected by this problem.Tree based algorithms were chosen because of their diversity and applicability in this type of prediction. These algorithms are Random Forest, support vector machine algorithm, and Logistic Regression algorithm. The method of preparation and selection of features and entering the mobile social network features had the biggest impact on the success of this model, since the value of AUC reached 93.301%. Random forest algorithm model achieved the best results in all measurements. The AUC value was 93.301%. The Support vector machine algorithm comes in the second place and the Logistic regression model came third. We have evaluated the models by fitting a new dataset related to different periods and without any proactive action from marketing, also gave the best result with 94% AUC. The decrease in result could be due to the non-stationary data model phenomenon, so the model needs training each period of time.

So this particular model is more useful for the telecom industries so that they can provide a reasonable and better structure to give the customers a good services so that they will not churn from their company which helps them to save their business.

6.2 FUTURE ENHANCEMENT

As future work, we can test different prediction algorithms (e.g., neural networks) to understand if it is possible to improve even more the prediction accuracy, and will extend the regression to other relevant metrics in the network, to verify the limits of what can be actually predicted in a cellular network. As this is the era of 5G there is so much of predictions needed regarding the 4G.By these predictions the telecom companies can understand the places they need to start 5G for the testing on human whether it is working or not.

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APPENDIX

A . PLAGIARISM REPORT

B. JOUNAL PAPER

C. SOURCE CODE

```
!pip install pymysql

import pandas as pd

import seaborn as

sns

import matplotlib.pyplot as plt

%matplotlib inline

from sqlalchemy import

create_engine db_host =

'18.136.56.185'

username = 'dm_team3'

user_pass =

'dm_team15119#' db_name =

'project_telecom'


conn                                                                      =
create_engine('mysql+pymysql://'+username+':'+user_pass+'@'+db_host+'/'+db_
name)

sql = 'select * from telecom_churn_data'

df = pd.read_sql(sql,conn)
```

```
df.columns=['State', 'Acc_len', 'Area_code', 'Phone', 'Int_plan',
    'Vmail_plan', 'Vmail_message', 'Day_min', 'Day_calls', 'Day_charge',
    'Eve_mins', 'Eve_calls', 'Eve_charge', 'Night_mins', 'Night_calls',
    'Night_charge', 'Int_mins', 'Int_calls', 'Int_charge', 'Custserv_calls',
    'Churn']
```

```
df.info()

df.head()

print(df.shape)

df.isnull().sum()

df.columns

#creating a list except the elements which are catergorical[state,Int_plan,Phone,Vmail_plan,Churn]

list = ['Acc_len', 'Area_code','Vmail_message', 'Day_min',
    'Day_calls', 'Day_charge','Eve_mins',

   'Eve_calls', 'Eve_charge', 'Night_mins', 'Night_calls',

   'Night_charge', 'Int_mins', 'Int_calls', 'Int_charge',  'Custserv_calls']



#ff['Acc_len']= pd.to_numeric(ff.Acc_len,errors='coerce')



df[list]= df[list].apply(pd.to_numeric,errors='coerce')

#dropping phone field -not required since it is

unique df.drop(columns='Phone',inplace=True)

print(df.shape)

#saving the file before proceeding further
```

```
df_bfre_enc=

df.to_csv('df_bfre_enc.csv',index=False) from

sklearn.preprocessing import LabelEncoder
```

```
list1=['Int_plan','Vmail_plan','Churn']

encoder = LabelEncoder()

for i in list1:

    df[i]=encoder.fit_transform(df[i])

df.State.value_counts()

df.State=pd.get_dummies(df.State,dtype='int64

') df.State.value_counts()

cleaned =

df.to_csv('cleaned_dataset.csv',index=False) import

pandas as pd

import matplotlib.pyplot as

plt import seaborn as sns

%matplotlib inline

#analysing data like

#GOAL: 2 classes Churners and non-Churners

#which features are the driving factors for

churn df=pd.read_csv('cleaned_dataset.csv')

df.head()
```

```
ff.groupby(['Churn']).std()

ff.groupby(['Churn']).mean()
```

gg=pd.read_csv('df_after_enc.csv')

gg.groupby('Area_code')['Churn'].value_counts()

df.head()

#checking the distribution of variables-normal distrubition(-1.96 to

+1.96) from scipy import stats

list=['Acc_len', 'Area_code','Vmail_message', 'Day_min', 'Day_calls','Day_charge','Eve_mins',

'Eve_calls', 'Eve_charge', 'Night_mins', 'Night_calls',

'Night_charge', 'Int_mins', 'Int_calls', 'Int_charge', 'Custserv_calls']

for i in list:

print('skewness and Kurtosis for '+ str(i)+': '+str(stats.skew(df[i]))+' and '+ str(stats.kurtosis(df[i])

#conclusion: we can see the Normality for International calls is not

statisfied sns.distplot([df.Acc_len])

#comparing the results there is not much difference

sns.boxplot(x='Churn',y='Acc_len',data=df,sym="")

sns.boxplot(x='Churn',y='Acc_len',data=df,sym="",hue='Int_plan')

#the mean and std of customer services calls for the churners is high compared to non-churner

sns.boxplot(x='Churn',y='Custserv_calls',data=df)

#we can see that churners call customer service

more

```
#checking with international plan now

sns.boxplot(x='Churn',y='Custserv_calls',data=df,hue='Int_plan',sym="")

sns.boxplot(x='Churn',y='Custserv_calls',data=df,hue='Vmail_plan')

#adding 3rd variable Internation plan for the above analysis and compare the churn
results

sns.boxplot(x='Churn',y='Custserv_calls',data=df,hue='Area_code',sym=""

) df.columns

sns.boxplot(x='Churn',y='Day_min',data=df,sym="")

sns.boxplot(x='Churn',y='Day_min',data=df,sym="")

sns.boxplot(x='Churn',y='Eve_mins',data=df,sym="")

sns.boxplot(x='Churn',y='Night_mins',data=df,sym="")

sns.boxplot(x='Churn',y='Night_mins',data=df,sym="")
```

```
import pandas as pd

import matplotlib.pyplot as

pltimportseabornassns

%matplotlib inline

#df= pd.read_csv('df_after_enc.csv')
ff=pd.read_csv('cleaned_dataset.csv')
```

```
ff.head()
```

```
#Scaling of the features

#Lets check the variance in magnitude of the features

ff.describe()

#to scale import standardizer

from sklearn.preprocessing import

StandardScaler #scaling on the quantitative data

list=['Acc_len','Vmail_message', 'Day_min', 'Day_calls', 'Day_charge',

     'Eve_mins', 'Eve_calls', 'Eve_charge', 'Night_mins', 'Night_calls',

     'Night_charge', 'Int_mins', 'Int_calls', 'Int_charge',  'Custserv_calls']



#list=['State','Acc_len',
                                    'Area_code','Int_plan','Vmail_plan','Vmail_message
', 'Day_min', 'Day_calls', 'Day_charge', 'Eve_mins',

    # 'Eve_calls', 'Eve_charge', 'Night_mins', 'Night_calls',

     'Night_charge', #'Int_mins', 'Int_calls', 'Int_charge', 'Custserv_calls']



ff[list] = StandardScaler().fit_transform(ff[list])



ff.info()
```

```
#scaling on the quantitative data
list=['Acc_len','Vmail_message',  'Day_min',  'Day_calls',  'Day_charge',
      'Eve_mins', 'Eve_calls', 'Eve_charge', 'Night_mins', 'Night_calls',
      'Night_charge',
```

```
        'Int_mins', 'Int_calls', 'Int_charge', 'Custserv_calls']



#list=['State','Acc_len',
                                'Area_code','Int_plan','Vmail_plan','Vmail_message
', 'Day_min', 'Day_calls', 'Day_charge', 'Eve_mins',

      # 'Eve_calls', 'Eve_charge', 'Night_mins', 'Night_calls',

        'Night_charge', #'Int_mins', 'Int_calls', 'Int_charge', 'Custserv_calls']



ff[list] = StandardScaler().fit_transform(ff[list])

ff.Churn=ff.Churn.astype('int64')

data_cor= ff.corr()

#FINDING THE CORRELATION bewtween the variables to drop the redundant
columns

data_cor= ff.corr()

data_cor.Churn.sort_values()

#saving file after only scaling

df_after_scale =ff.to_csv('df_after_scale.csv',index=False)

conclusion we can see that day mins-day charge / eve_mins-eve charge / night mins
-night charge/ Int mins and Int charge are highly correlated

# we can add new features as well -TBD LATER
```

```
df =

ff.drop(['Day_charge','Eve_charge','Night_charge','Int_charge'],axis=1)

df.head()
```

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4617 entries, 0 to 4616
Data columns (total 16 columns):
State             4617 non-null int64
Acc_len           4617 non-null float64
Area_code         4617 non-null int64
Int_plan          4617 non-null int64
Vmail_plan        4617 non-null int64
Vmail_message     4617 non-null float64
Day_min           4617 non-null float64
Day_calls         4617 non-null float64
Eve_mins          4617 non-null float64
Eve_calls         4617 non-null float64
Night_mins   4617   non-null   float64
Night_calls   4617   non-null   float64
Int_mins          4617 non-null float64
Int_calls         4617 non-null float64
Custserv_calls    4617 non-null float64
Churn             4617 non-null int64
dtypes: float64(11), int64(5)
memory usage: 577.2 KB
```

df=df.to_csv('df_scale_corr.csv',index=False)

saved file after scaling and dropping highly correalted

variables. import pandas as pd

import matplotlib.pyplot as

pltimportseabornassns

%matplotlib inline

since the output variable is Goal- churn or no churn TRIAL AND ERROR between logistic regression ,SVM ,Randomforest

```
from sklearn.linear_model import

LogisticRegression from sklearn.svm import SVC

from sklearn.ensemble import

RandomForestClassifier from sklearn.model_selection

import train_test_split from collections import Counter

from imblearn.over_sampling import SMOTE

from                              sklearn.metrics
                                  import
accuracy_score,classification_report,confusion_matrix,recall_score,precision_sc
o re

df=

pd.read_csv('df_after_scale.csv') X

= df.drop('Churn',axis=1)

X.head()

y=df.Churn

X_train,X_test,y_train,y_test                                                    =
train_test_split(X,y,test_size=0.3,random_state=10)

print(X_train.shape)

print(X_test.shape)

print(Counter(y_train))
```

```
print(Counter(y_test))

#### SMOTING REQUIRED -since the ratio of churners and non-churners is not equal

sm = SMOTE()
```

```
X_train_sm,y_train_sm = sm.fit_sample(X_train,y_train)

# now the chuners and non chuners ratio is same in the output

print(Counter(y_train_sm))

print(Counter(y_test))

clf=LogisticRegression()

clf.fit(X_train_sm,y_train_sm)

y_pred=clf.predict(X_test)

#checking accuracy score

accuracy_score(y_test,y_pred)

recall_score(y_test,y_pred)

precision_score(y_test,y_pred)

confusion_matrix(y_test,y_pred)

pd.crosstab(y_test,y_pred)

print(classification_report(y_test,y_pred))
```

For the LR though the recall is 78% the precision score is very less

Support vector Machine

```
svc= SVC()

svc.fit(X_train_sm,y_train_sm)

y_pred_svc= svc.predict(X_test)
```

```
accuracy_score(y_test,y_pred_svc)
```

print(classification_report(y_test,y_pred_svc))

confusion_matrix(y_test,y_pred_svc)

pd.crosstab(y_test,y_pred)

Random forest

model = RandomForestClassifier(random_state=5,n_estimators=100,max_depth=15)

model.fit(X_train_sm,y_train_sm)

model.feature_importances_

y_pred_rf= model.predict(X_test)

print(classification_report(y_test,y_pred_rf))

accuracy_score(y_test,y_pred_rf)

confusion_matrix(y_test,y_pred_rf)

Random forest gives best results for all the metrics as compared to LR and SVM

import pandas as pd

import numpy as np

from sklearn.linear_model import LogisticRegression

from sklearn.ensemble import

RandomForestClassifier from imblearn.over_sampling

import SMOTE

from collections import Counter

```
from sklearn.model_selection import GridSearchCV,RandomizedSearchCV

from sklearn.metrics import classification_report,confusion_matrix,accuracy_score,recall_score,precision_score,roc_curve,roc_auc_score



from sklearn.model_selection import train_test_split

import matplotlib.pyplot as plt

df= pd.read_csv('cleaned_dataset.csv')

df.head()

df.columns

df = df.astype('float')

df.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 4617 entries, 0 to 4616

Data columns (total 20 columns):

 # Column         Non-Null Count Dtype

 --  ----         ---------  -----

 0   State        4617 non-null float64

 1   Acc_len      4617 non-null float64

 2   Area_code    4617 non-null float64

 3   Int_plan     4617 non-null float64
```

4 Vmail_plan 4617 non-null float64

5 Vmail_message 4617 non-null float64

6 Day_min 4617 non-null float64

7 Day_calls 4617 non-null float64

8 Day_charge 4617 non-null float64

9 Eve_mins 4617 non-null float64

10 Eve_calls 4617 non-null float64

11 Eve_charge 4617 non-null float64

12 Night_mins 4617 non-null float64

13 Night_calls 4617 non-null float64

14 Night_charge 4617 non-null float64

15 Int_mins 4617 non-null float64

16 Int_calls 4617 non-null float64

17 Int_charge 4617 non-null float64

18 Custserv_calls 4617 non-null float64

19 Churn 4617 non-null float64

dtypes: float64(20) memory

usage: 721.5 KB X =

df.drop('Churn',axis=1)

y=df.Churn

```
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.3,random_state=10)

print(X_train.shape)

print(X_test.shape)

print(Counter(y_train))

print(Counter(y_test))

sm = SMOTE()

X_train_sm,y_train_sm = sm.fit_sample(X_train,y_train)

sm = SMOTE()

X_train_sm,y_train_sm = sm.fit_sample(X_train,y_train)

# now the chuners and non chuners ratio is same in the output

print(Counter(y_train_sm))

print(Counter(y_test))

model = RandomForestClassifier(n_estimators=9,

 max_features='sqrt',

 max_depth=9,

 criterion='gini',

 bootstrap=False)



model.fit(X_train_sm,y_train_sm)

y_pred_rf= model.predict(X_test)
```

print(classification_report(y_test,y_pred_rf))

got a recall of 82% and precison of 80% at a particular trial

accuracy_score(y_test,y_pred_rf)

confusion_matrix(y_test,y_pred_rf)

Grid Search Method

params= {'n_estimators':np.arange(1,10),

'criterion': ['entropy','gini'],

'max_features':['auto','sqrt','log2'],

'bootstrap':[True,False]

}

model_cv=RandomizedSearchCV(RandomForestClassifier(),params)

#model_cv =GridSearchCV(RandomForestClassifier(),params)

model_cv.fit(X_train_sm,y_train_sm)

model_cv.best_params_

model_cv.best_estimator_

ROC CURVE

#checking for the ROC - CHECK FOR EFFICIENCY OF THE MODEL

model.predict_proba(X_test)

#EXTRACTING ONLY the 2nd column values as it the predict proba values for the churners

```
y_pred_prob=model.predict_proba(X_test)[:,1]

print(y_pred_prob)

fpr, tpr , thresholds = roc_curve(y_test,y_pred_prob)

plt.plot(fpr,tpr)

plt.xlabel("False positive rate")

plt.ylabel("True positive rate")

plt.plot([0,1],[0,1],"--")

plt.show()

# checking the area under the curve - for a default random model AUC - 0.5

#So, we make sure we get better AUC score for a better performing model

auc = roc_auc_score(y_test,y_pred_prob)

print(auc)
```

FROM THE Above result we can see the area under the curve is 90% which indicates the model is pretty good for the predictions further

```
importances= model.feature_importances_

sort=np.argsort(importances)

label=X.columns[sort]

X.shape

ss= range(X.shape[1])

print(ss)

plt.barh(ss,importances[sort],tick_label=label)
```

The above figure shows the feature importances for the particular model.and which features drive churn the most

import joblib

joblib.dump(model,'Telecom_trained_model.ml')

#to load the trained model

model=joblib.load('Telecom_trained_model.ml')